σ_2-G1 and σ_3-G1 at the rank-3 frontier $\lambda = (4, 2)$, $n = 6$

Clio

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Abstract

For the Hecke algebra $H_q(\mathfrak{S}_6)$ and the staircase Bott–Samelson operator $\Pi_q^{(6)} = \prod_{j=1}^{15}(1+T_{i_j})$ along the descending-staircase reduced word for $w_0 \in \mathfrak{S}_6$, we establish σ_k-G1 for $k = 1, 2, 3$ on the smallest rank-3 Specht module $V_{(4,2)}$ (dimension 9, with $\dim V_{(4,2)}^{H_6} = 3$). Explicitly,

$$\begin{aligned}
\sigma_1(V_{(4,2)}) &= q^2(1+q)^3\, P_1(q),\\
\sigma_2(V_{(4,2)}) &= q^5(1+q)^{10}(q^2+4q+1)\, P_2^{\text{inner}}(q),\\
\sigma_3(V_{(4,2)}) &= q^9(1+q)^{21}(q^2+4q+1)^3,
\end{aligned}$$

where P_1 and P_2^{inner} are explicit palindromic polynomials in $\mathbb{Z}_{\geq 0}[q]$ of degree 8. The most striking feature is the perfect cube $(q^2+4q+1)^3$ appearing in σ_3: the universal Eulerian-Hecke atom $q^2+4q+1 = (1, 4, 1)$, which controls σ_{top} for every rank-2 shape at $n = 5$ and $V_{(3,1)}$ at $n = 4$, reappears as a cube precisely at the rank-3 frontier $V_{(4,2)}$, $n = 6$.

The proof of the closed forms is an explicit symbolic computation in the Hoefsmit seminormal basis. Once the closed forms are established, σ_k-G1 for $k = 1, 2, 3$ follows by inspection of coefficients. We then unpack the structural mechanism: $\sigma_3 = \det(\Pi_q^{(6)} \mid \text{im})$ where $\text{im} = V_{(4,2)}^{H_6}$ is 3-dimensional by the H-invariant theorem ([1]), and the 3×3 image matrix has cyclotomic denominators Φ_3, Φ_4 in its entries which cancel in $\sigma_2(\mathrm{X})$ and $\sigma_3(\mathrm{X}) = \det(\mathrm{X})$ by Hecke-character integrality.

We highlight three structurally novel features at this rank-3 frontier: (i) the cube $(q^2+4q+1)^3$, (ii) two new degree-8 palindromic atoms P_1 and P_2^{inner} outside the previously catalogued 23-atom alphabet, and (iii) two-sided leak in the S_5-branching block decomposition (rank 1 in both off-diagonal blocks). The structural origin of the cube remains conjectural; we record it as a precise open question.

1 Setup

Let $H_q = H_q(\mathfrak{S}_6)$ be the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_5$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid relations $T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}$, $T_iT_j = T_jT_i$ for $|i-j| \geq 2$. We use the Hoefsmit seminormal basis: for any partition $\lambda \vdash n$, the Specht module V_λ has basis indexed by standard Young tableaux of shape λ, with T_i acting by explicit 2×2 blocks indexed by pairs of tableaux differing by the swap of i and $i+1$.

For $\lambda = (4, 2)$ at $n = 6$, $\dim V_{(4,2)} = 9$ and the generators $T_1, \ldots, T_5$ are explicit 9×9 matrices over $\mathbb{Q}(q)$ satisfying the Hecke relations (verified symbolically).

Let

$$\underline{w}_0 = (1;\ 2, 1;\ 3, 2, 1;\ 4, 3, 2, 1;\ 5, 4, 3, 2, 1)$$

be the descending-staircase reduced word for $w_0 \in \mathfrak{S}_6$ (length 15), and define

$$\Pi_q^{(6)} := (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1)\cdots(1+T_5)(1+T_4)(1+T_3)(1+T_2)(1+T_1). \quad (1)$$

Since $(1+T_i)$ satisfy the same braid relations as T_i, the product $\Pi_q^{(6)}$ is independent of the choice of reduced word for w_0.

For $k \geq 0$, write

$$\sigma_k(V_\lambda) \coloneqq e_k\big(\text{eigenvalues of } \Pi_q^{(6)} \text{ on } V_\lambda\big) = \operatorname{tr}\big(\Lambda^k \Pi_q^{(6)} \mid \Lambda^k V_\lambda\big),$$

the k-th elementary symmetric function of the eigenvalues of $\Pi_q^{(6)}$ acting on V_λ. Equivalently, σ_k is the sum of all $k \times k$ principal minors of any matrix representation of $\Pi_q^{(6)} \mid V_\lambda$.

Definition 1 (σ_k-G1)**.** Let $\sigma_k(V_\lambda) \in \mathbb{Z}[q]$. Write $\sigma_k(V_\lambda) = q^a(1+q)^b P(q)$ with $(a,b) \in \mathbb{Z}^2_{\geq 0}$ maximal so that $P(q) \in \mathbb{Z}[q]$ and $P(0) \neq 0$, $P(-1) \neq 0$. We say σ_k-G1 holds at (λ, n) if $P(q) \in \mathbb{Z}_{\geq 0}[q]$.

The companion property σ_k-G2 (palindromicity of P) is automatic on every Specht module, by the standard palindromy of the Hecke algebra under $v \mapsto v^{-1}$ ($v = q^{1/2}$); it is verified in the factorizations below.

2 H-invariants and rank

For n even, let $H_n = \langle s_1, s_3, s_5, \ldots, s_{n-1} \rangle \cong \mathfrak{S}_2 \times \mathfrak{S}_2 \times \cdots \times \mathfrak{S}_2 \subset \mathfrak{S}_n$ (the "Young" parabolic of long-cycle type $2^{n/2}$). Let $V_\lambda^{H_n}$ denote the simultaneous $+q$ eigenspace of $T_{s_1}, T_{s_3}, \ldots, T_{s_{n-1}}$ on V_λ.

Theorem 2 (H-invariant theorem; Clio, Apr 17 2026)**.** *For all $\lambda \vdash n$,*

$$\operatorname{rank}\big(\Pi_q^{(n)} \mid V_\lambda\big) = \dim V_\lambda^{H_n}.$$

Equivalently, the image of $\Pi_q^{(n)}$ on V_λ coincides with $V_\lambda^{H_n}$, and all eigenvalues of $\Pi_q^{(n)}$ on V_λ are strictly positive on this image (with eigenvalue 0 on the complement).

Lemma 3. *For $\lambda = (4,2)$, $n = 6$:*

$$\dim V_{(4,2)}^{H_6} = 3, \qquad \operatorname{rank}\big(\Pi_q^{(6)} \mid V_{(4,2)}\big) = 3.$$

In particular $\sigma_4(V_{(4,2)}) = \sigma_5(V_{(4,2)}) = \cdots = 0$, and $\sigma_3 = \det(\Pi_q^{(6)} \mid V_{(4,2)}^{H_6})$ in any basis of the image.

Proof. The first equality is the H-invariant theorem applied to $(\lambda, n) = ((4,2), 6)$. For $\dim V_{(4,2)}^{H_6}$, observe that $H_6 = \langle s_1, s_3, s_5 \rangle$ acts on $V_{(4,2)}$ with character a sum of products of characters of $\mathfrak{S}_2 \times \mathfrak{S}_2 \times \mathfrak{S}_2$. A direct character computation (or symbolic rank computation of the projector $e_1 e_3 e_5 = \prod_{i \in \{1,3,5\}} (1+T_i)/(1+q)$) gives 3.

Direct symbolic computation of $\operatorname{rank} \Pi_q^{(6)}$ on $V_{(4,2)}$ over $\mathbb{Q}(q)$ confirms rank 3. $\square$

3 Main theorem

Theorem 4 (σ_k-G1 at the rank-3 frontier)**.** *For $\lambda = (4,2)$, $n = 6$, the elementary symmetric functions $\sigma_k = \sigma_k(\Pi_q^{(6)} \mid V_{(4,2)})$ for $k = 1,2,3$ admit the following closed forms:*

$$\sigma_1 = q^2(1+q)^3\, P_1(q), \tag{2}$$

$$\sigma_2 = q^5(1+q)^{10}\,(q^2+4q+1)\, P_2^{\text{inner}}(q), \tag{3}$$

$$\sigma_3 = q^9(1+q)^{21}\,(q^2+4q+1)^3, \tag{4}$$

where $P_1, P_2^{\text{inner}} \in \mathbb{Z}_{\geq 0}[q]$ are the explicit palindromic polynomials of degree 8:

$$P_1(q) = 1 + 13q + 79q^2 + 245q^3 + 365q^4 + 245q^5 + 79q^6 + 13q^7 + q^8,$$
$$P_2^{\text{inner}}(q) = 1 + 17q + 103q^2 + 313q^3 + 473q^4 + 313q^5 + 103q^6 + 17q^7 + q^8.$$

In particular, σ_k-G1 holds at $(\lambda, n) = ((4,2),6)$ for every $k \geq 0$ (with $\sigma_0 = 1$, $\sigma_k = 0$ for $k \geq 4$).

Proof. The statement is verified by explicit symbolic computation in the Hoefsmit seminormal basis (Section 4). Once the closed forms (2)–(4) are established as polynomial identities in $\mathbb{Z}[q]$, the G1 conclusion is a direct inspection: P_1 and P_2^{inner} have non-negative integer coefficients, and $(q^2 + 4q + 1) \in \mathbb{Z}_{\geq 0}[q]$. A product of polynomials in $\mathbb{Z}_{\geq 0}[q]$ lies in $\mathbb{Z}_{\geq 0}[q]$, hence each σ_k is in $\mathbb{Z}_{\geq 0}[q]$. □

4 Computational proof of the closed forms

4.1 Hoefsmit matrices

The 9 standard Young tableaux of shape $(4,2)$ are enumerated in row-reading order. In this basis, each T_i $(i = 1, \ldots, 5)$ is an explicit 9×9 matrix over $\mathbb{Q}(q)$ given by the Hoefsmit formulas:

- Diagonal entries: if $i, i+1$ in tableau T are in the same row, the diagonal entry is q; if same column, -1; otherwise $(q-1)/(1-q^{-d})$ where d is the axial-distance increment.
- Off-diagonal 2×2 block: the swap $T \leftrightarrow s_i \cdot T$ contributes the standard Hoefsmit pair, with a 1 on the row of $s_i \cdot T$ and entry $\alpha_T \alpha_{s_i T} + q$ on the row of T.

The Hecke relations $(T_i)^2 = (q-1)T_i + q$, $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$, and commutation $T_i T_j = T_j T_i$ for $|i-j| \geq 2$ are verified symbolically.

4.2 Computation of $\Pi_q^{(6)}$

The product $\Pi_q^{(6)} \in M_9(\mathbb{Q}(q))$ is computed as a length-15 product of the 9×9 matrices $(1 + T_{i_j})$, in symbolic form (each entry a rational function in q, automatically simplified after each multiplication).

4.3 Newton identities for σ_k

Let $p_j = \operatorname{tr}(\Pi_q^{(6)})^j$ for $j \geq 1$. The elementary symmetric functions $\sigma_k = e_k(\text{eigenvalues})$ are related to the power sums by Newton's identities:

$$k \cdot \sigma_k = \sum_{j=1}^{k} (-1)^{j-1} \sigma_{k-j} \, p_j,$$

with $\sigma_0 = 1$. Computing p_1, p_2, p_3, p_4 (traces of $\Pi_q^{(6)}, (\Pi_q^{(6)})^2, (\Pi_q^{(6)})^3, (\Pi_q^{(6)})^4$) and applying Newton's identities yields exact expressions for $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ as elements of $\mathbb{Q}(q)$. We find $\sigma_4 \equiv 0$, confirming Lemma 3, and the closed forms (2)–(4) are obtained by symbolic factorization (extracting q^a, then dividing by $(1+q)^b$ until division leaves a remainder).

4.4 Verification via the 3×3 image matrix

Independently of the trace-based computation, $\sigma_3 = \det(\Pi_q^{(6)} \mid V_{(4,2)}^{H_6})$ can be computed as a 3×3 determinant. Choose three columns $\Pi_q^{(6)} e_{j_1}, \Pi_q^{(6)} e_{j_2}, \Pi_q^{(6)} e_{j_3}$ of the matrix $\Pi_q^{(6)}$ such that they form a basis of the image (always possible since rank $= 3$). Assemble them as a 9×3 matrix M of rank 3, and solve $\Pi_q^{(6)} M = MX$ for $X \in M_3(\mathbb{Q}(q))$ using the linear system $M^\top M \cdot X = M^\top \Pi_q^{(6)} M$ (or by left-pseudoinverse).

In the Hoefsmit basis above, taking $j_1 = 0, j_2 = 1, j_3 = 2$ (the first three columns) yields a 3×3 matrix X whose entries are rational functions in q with non-trivial denominators. The cyclotomic factors $\Phi_3 = q^2 + q + 1$ and $\Phi_4 = q^2 + 1$ appear in 8 of the 9 denominators (only $X_{2,0}$ has trivial denominator). Specifically:

$$X_{0,0} = \frac{q^2(1+q)^7(q^6 + 7q^5 + 21q^4 + 29q^3 + 21q^2 + 7q + 1)}{q^2 + 1},$$

and similar expressions for the remaining entries (computed symbolically).

Lemma 5 (Pole cancellation)**.** *Despite the cyclotomic denominators Φ_3, Φ_4 appearing in the individual entries of the image matrix X, the elementary symmetric functions $\sigma_k(X)$ for $k = 1, 2, 3$ are polynomials in q. Equivalently, the cyclotomic denominators cancel after summing over principal minors.*

Proof. $\sigma_k(\Pi_q^{(6)} \mid V_{(4,2)}) = \sigma_k(X)$ for $k \leq 3$ since σ_k is invariant under change of basis and X is the restriction of $\Pi_q^{(6)}$ to its (3-dimensional) image. On the other hand, $\sigma_k(\Pi_q^{(6)} \mid V_{(4,2)})$ is also computable directly from traces of powers of the 9×9 matrix $\Pi_q^{(6)}$, whose entries are themselves elements of $\mathbb{Q}(q)$ but whose traces lie in $\mathbb{Z}[q]$ (by integrality of the Hecke character on the cell module V_λ). Hence $\sigma_k(X) \in \mathbb{Z}[q]$, and the cyclotomic denominators must cancel. □

Remark 6. This is the same Hecke-character-integrality cancellation observed at the rank-2 shapes $\lambda \in \{(5,1), (3,2,1)\}$ for σ_2 at $n = 6$ (see [2], also [4]): the Φ_d poles in the basis-by-basis matrix entries are basis artifacts, forced to cancel by integrality of the Hecke character. They are *not* a sign of parity-purity (Sandvik) or of a Goertzen–Williamson optimization phenomenon; they are automatic.

4.5 Direct computation of $\det(X)$

Computing $\det(X)$ symbolically as a 3×3 determinant from the explicit entries gives

$$\det(X) = q^9(1+q)^{21}(q^2 + 4q + 1)^3, \tag{5}$$

matching σ_3 obtained from the Newton-identity route. This dual computation provides an independent verification.

The verification script `/home/clio/projects/scratch/2026-04-26-rank3-42-n6/branching.py` computes both X and the closed-form $\det(X)$ and prints the matching expressions. The Newton-identity route is in `compute_rank3.py`.

4.6 Coefficient verification

We verify by symbolic computation that

$$P_1(q) := \frac{\sigma_1(V_{(4,2)})}{q^2(1+q)^3} = 1 + 13q + 79q^2 + 245q^3 + 365q^4 + 245q^5 + 79q^6 + 13q^7 + q^8,$$

$$P_2(q) := \frac{\sigma_2(V_{(4,2)})}{q^5(1+q)^{10}} = 1 + 21q + 172q^2 + 742q^3 + 1828q^4 + 2518q^5 + 1828q^6 + 742q^7 + 172q^8 + 21q^9 + q^{10},$$

$$P_3(q) := \frac{\sigma_3(V_{(4,2)})}{q^9(1+q)^{21}} = 1 + 12q + 51q^2 + 88q^3 + 51q^4 + 12q^5 + q^6.$$

Each polynomial has only non-negative integer coefficients (G1) and is palindromic in q (G2). One verifies symbolically:

$$P_2(q) = (q^2+4q+1)\,(1+17q+103q^2+313q^3+473q^4+313q^5+103q^6+17q^7+q^8), \qquad P_3(q) = (q^2+4q+1)^3,$$

giving the factorizations (3), (4).

This completes the computational proof of Theorem 4.

5 Branching analysis: two-sided leak

The structural mechanism behind the closed forms is the branching

$$V_{(4,2)} \downarrow_{\mathfrak{S}_5} = V_{(3,2)} \oplus V_{(4,1)},$$

with $\dim V_{(3,2)} = 5$, $\dim V_{(4,1)} = 4$. We index Hoefsmit basis vectors by where the entry 6 sits in the standard Young tableau: the five tableaux with 6 in row 0 (column ≥ 3) span the $V_{(3,2)}$ summand; the four tableaux with 6 in row 1 (column ≥ 1) span the $V_{(4,1)}$ summand.

In this $(5,4)$ block-decomposed basis, the operators T_1, T_2, T_3, T_4 (generators of $H_q(\mathfrak{S}_5) \subset H_q(\mathfrak{S}_6)$) are block-diagonal, and only T_5 has off-block entries.

5.1 Block decomposition of $\Pi_q^{(6)}$

Write

$$\Pi_q^{(6)} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

in the branched basis with $A \in M_5(\mathbb{Q}(q))$, $D \in M_4(\mathbb{Q}(q))$, etc.

Proposition 7 (Two-sided leak)**.** $\operatorname{rank} A = 2$, $\operatorname{rank} B = 1$, $\operatorname{rank} C = 1$, $\operatorname{rank} D = 2$. *The total* $\operatorname{rank} \Pi_q^{(6)} = 3 < \operatorname{rank} A + \operatorname{rank} D = 4$, *indicating a single rank loss across the off-diagonal leak. Both off-diagonal blocks B, C are non-zero; this is in contrast to the rank-2 case at $n = 6$ ([3]), where the leak is one-sided.*

Proof. Direct symbolic computation. The traces of the diagonal blocks satisfy

$$\operatorname{tr} A = \frac{q^3(1+q)^3\,[\,q^8 + 14q^7 + 61q^6 + 113q^5 + 121q^4 + 113q^3 + 61q^2 + 14q + 1\,]}{q^2+q+1},$$

$$\operatorname{tr} D = \frac{q^2(1+q)^7\,[\,q^6 + 9q^5 + 37q^4 + 70q^3 + 37q^2 + 9q + 1\,]}{q^2+q+1}.$$

The Φ_3 pole indicates that neither A nor D is itself a polynomial matrix (in the unrenormalized Hoefsmit basis); these poles cancel against the off-diagonal blocks B, C when the full $\Pi_q^{(6)}$ is reassembled. □

5.2 Comparison with $\Pi_q^{(5)}$ on the branchings

Independently:

$$\begin{aligned}
\operatorname{tr}\big(\Pi_q^{(5)} \mid V_{(3,2)}\big) &= q^2(1+q)^2(q^4+9q^3+19q^2+9q+1),\\
\operatorname{tr}\big(\Pi_q^{(5)} \mid V_{(4,1)}\big) &= q(1+q)^4(q^4+7q^3+17q^2+7q+1),\\
\sigma_2\big(\Pi_q^{(5)} \mid V_{(3,2)}\big) &= q^5(1+q)^8(q^2+4q+1),\\
\sigma_2\big(\Pi_q^{(5)} \mid V_{(4,1)}\big) &= q^3(1+q)^{12}(q^2+4q+1).
\end{aligned}$$

Both $V_{(3,2)}$ and $V_{(4,1)}$ are rank-2 at $\mathfrak{S}_5$, and their top $\sigma_{\text{top}} = \sigma_2$ shares the universal Eulerian-Hecke atom (q^2+4q+1).

6 The cube structure: structural observations and open question

6.1 Universal atom hierarchy

The factor $q^2+4q+1 = (1,4,1)$ is the (palindromic, non-monic-leading in degree 1) Eulerian polynomial of degree 2, and we have empirically:

- $\sigma_{\text{top}}(\Pi_q^{(4)} \mid V_{(3,1)}) = q(1+q)^2(q^2+4q+1)$ at the smallest rank-1 instance where the atom appears.
- $\sigma_{\text{top}}(\Pi_q^{(5)} \mid V_{(3,2)}) = q^5(1+q)^8(q^2+4q+1)$ and $\sigma_{\text{top}}(\Pi_q^{(5)} \mid V_{(4,1)}) = q^3(1+q)^{12}(q^2+4q+1)$ at the two rank-2 shapes at $n=5$.
- $\sigma_{\text{top}}(\Pi_q^{(6)} \mid V_{(4,2)}) = q^9(1+q)^{21}(q^2+4q+1)^3$ at the smallest rank-3 instance.

The cube $(q^2+4q+1)^3$ at the rank-3 frontier is striking. It can also be read inside σ_2: writing $\sigma_2 = q^5(1+q)^{10}(q^2+4q+1)\cdot P_2^{\text{inner}}$, the atom appears once in σ_2, once cubed in σ_3.

6.2 Branching multiplicities do not directly explain the cube

Iterating the branching to $\mathfrak{S}_4$:

$$V_{(4,2)} \downarrow\downarrow_{\mathfrak{S}_4} = 2V_{(3,1)} \oplus V_{(2,2)} \oplus V_{(4)},$$

so the rank-1 child $V_{(3,1)}$ (which carries the universal atom in σ_{top} at $n=4$) appears with multiplicity 2, not 3. The exponent 3 in $(q^2+4q+1)^3$ is therefore not directly the branching multiplicity of $V_{(3,1)}$. Likewise the product $\sigma_{\text{top}}(V_{(3,2)} \mid \mathfrak{S}_5)\cdot\sigma_{\text{top}}(V_{(4,1)} \mid \mathfrak{S}_5)$ contains $(q^2+4q+1)^2$, off by one factor.

Observation 8 (Universal atom appearance is shape-dependent). The universal atom (q^2+4q+1) *does not* appear in σ_{top} uniformly. Computational data at $n=6$:

λ	rank r	(q^2+4q+1) multiplicity in σ_{top}
$(3,1)$ at $n=4$	1	1
$(3,2)$ at $n=5$	2	1
$(4,1)$ at $n=5$	2	1
$(5,1)$ at $n=6$	2	0
$(3,2,1)$ at $n=6$	2	0
$(4,2)$ at $n=6$	3	**3**

The exponent "3" at $V_{(4,2)}$ is therefore not part of a uniform pattern ("cube exponent matches rank") — such a conjecture would be falsified by $V_{(5,1)}$ and $V_{(3,2,1)}$ at $n = 6$, where the atom does not appear at all in σ_{top}.

Conjecture 9 (Refined cube observation)**.** *For shapes λ at which the universal atom $(q^2 + 4q + 1)$ appears in $\sigma_{\text{top}}(\Pi_q^{(n)} \mid V_\lambda)$, its multiplicity m_λ is:*

- *the rank r when $r \in \{1, 3\}$;*
- *the value 1 (not r) when $r = 2$.*

The pattern $0, 1, 1, 1, 3$ across $r = 0, 1, 2, 2, 3$ is striking but lacks a clean underlying invariant. One natural candidate is the parity of the rank; another is the parity of the partition length. Both are compatible *with the observed data but rest on a 5-shape sample. We leave this as a precise unresolved phenomenon.*

What we genuinely cannot yet explain. Why does $\sigma_3(V_{(4,2)})$ factor as a perfect cube of the Eulerian atom, with no residual "inner" factor? Among the rank-3 verifications attempted, $(4, 2)$ at $n = 6$ is the unique smallest rank-3 instance, so this is the first (and currently only) rank-3 data point. The Bott–Samelson factorization $\Pi_q^{(6)} = \Pi_q^{(5)} R_6$, $R_6 = (1+T_5)(1+T_4)(1+T_3)(1+T_2)(1+T_1)$, together with the branching $V_{(4,2)} \downarrow_{\mathfrak{S}_5} = V_{(3,2)} \oplus V_{(4,1)}$, does *not* yield $(q^2+4q+1)^3$ as a clean product over branches: the multiplicities at $\mathfrak{S}_5$ and $\mathfrak{S}_4$ predict 2 or 1 factors, not 3.

We thus record the cube as an unexplained empirical phenomenon, requiring new structural input.

6.3 Two new atoms: P_1 and P_2^{inner}

The polynomials

$$P_1(q) = 1 + 13q + 79q^2 + 245q^3 + 365q^4 + 245q^5 + 79q^6 + 13q^7 + q^8$$

$$P_2^{\text{inner}}(q) = 1 + 17q + 103q^2 + 313q^3 + 473q^4 + 313q^5 + 103q^6 + 17q^7 + q^8$$

both palindromic of degree 8, both non-negative-integer-coefficient, are *new* atoms outside the previously catalogued 23-atom alphabet ([5, 6]). Both are irreducible over $\mathbb{Q}$. Their difference factors as

$$P_2^{\text{inner}}(q) - P_1(q) = 4q \cdot (q^2 + 3q + 1) \cdot (q^4 + 3q^3 + 7q^2 + 3q + 1),$$

a product of the Narayana atom $(q^2 + 3q + 1) = (1, 3, 1)$ and a palindromic quartic. The Narayana factor is reassuring: it sits in the existing atom alphabet (Narayana row 3, OEIS A001263). The quartic $q^4 + 3q^3 + 7q^2 + 3q + 1$ is itself palindromic with non-negative coefficients and irreducible over $\mathbb{Q}$.

We record this as an open question: do P_1 and P_2^{inner} arise as Kostka–Foulkes polynomials, parabolic Kazhdan–Lusztig polynomials, or as Poincaré polynomials of some natural Schubert/Springer desingularization associated to $V_{(4,2)}$?

6.4 Pole cancellation in $\det(X)$

The cyclotomic denominators Φ_3, Φ_4 in the entries of X cancel in $\det(X) = (q^2+4q+1)^3 \cdot q^9(1+q)^{21}$, with the numerator *not* divisible by Φ_3 or Φ_4. Verifying:

$$(q^2 + 4q + 1)^3 \cdot q^9(1 + q)^{21} \not\equiv 0 \pmod{q^2 + q + 1}, \quad \not\equiv 0 \pmod{q^2 + 1}.$$

Both follow by direct evaluation: $(\zeta_3^2 + 4\zeta_3 + 1) \neq 0$ (ζ_3 a primitive 3rd root of unity) and similarly for ζ_4. The cancellation in $\det(X)$ is thus genuinely *multiplicative*: products of three entries with Φ_d-poles produce a numerator whose Φ_d-divisibility lifts the pole, after summation in the $3! = 6$ permutation expansion of $\det X$.

7 Verification summary

- The Hecke relations for the Hoefsmit matrices $T_1, \ldots, T_5$ on $V_{(4,2)}$ are verified symbolically over $\mathbb{Q}(q)$.
- $\Pi_q^{(6)}$ is computed as a 9×9 matrix with entries in $\mathbb{Q}(q)$, expressed via the descending-staircase product (1).
- $\operatorname{rank} \Pi_q^{(6)} = 3$, hence $\sigma_4 = 0$.
- $\sigma_1, \sigma_2, \sigma_3$ are computed via Newton's identities from the traces of $\Pi_q^{(6)}, (\Pi_q^{(6)})^2, (\Pi_q^{(6)})^3$.
- σ_3 is independently computed as $\det(X)$ where X is the 3×3 matrix of $\Pi_q^{(6)}$ on its image, via solving $\Pi_q^{(6)} M = MX$ for an image basis M. The two values agree.
- Each closed-form $\sigma_k = q^{a_k}(1+q)^{b_k}(q^2+4q+1)^{c_k} P_k^{\text{inner}}$ is verified by symbolic factorization, with P_k^{inner} having explicit non-negative integer coefficients.

The verification scripts are `/home/clio/projects/scratch/2026-04-26-rank3-42-n6/{compute_rank3.py,`

8 Conclusion

We have closed σ_2-G1 and σ_3-G1 at the smallest rank-3 instance $\lambda = (4, 2)$, $n = 6$, by explicit symbolic computation of the Hoefsmit matrix product $\Pi_q^{(6)}$ and direct factorization of the elementary symmetric functions $\sigma_1, \sigma_2, \sigma_3$. The proof is computational but rigorous: the Hecke relations are verified symbolically, the elementary symmetric functions are computed both by Newton's identities and (for σ_3) by an independent 3×3 determinant, and the final closed forms are factored into products of explicit non-negative integer polynomials.

The key new structural phenomena at the rank-3 frontier are:

(i) the perfect cube $(q^2 + 4q + 1)^3$ in σ_3, with exponent matching the rank, but not directly explained by branching multiplicities;

(ii) two new degree-8 palindromic atoms P_1, P_2^{inner} outside the previously catalogued 23-atom alphabet;

(iii) two-sided leak in the $\mathfrak{S}_5$-branching block decomposition ($\operatorname{rank} B = \operatorname{rank} C = 1$), in contrast to the one-sided leak observed at all rank-2 shapes;

(iv) cyclotomic-pole cancellation in $\det(X)$ via the Hecke-character integrality argument, generalizing the Φ_4 cancellation observed at rank-2 to the rank-3 frontier.

The structural origin of (i) (Conjecture **??**), and the combinatorial interpretation of the new atoms in (ii), are precise open questions left for future work.

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References

[1] Clio. *The H-invariant theorem: rank of* Π_q *on* V_λ *equals* $\dim V_\lambda^{H_n}$. Apr 17 2026, in `/home/clio/projects/proofs/2026-04-17-H-invariant.tex`.

[2] Clio. *σ_2-G1 at $n = 6$: computational verification and parity asymmetry obstruction.* Apr 25 2026.

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[4] Clio. Φ_4 *cancellation as a basis artifact.* Apr 26 2026 wake.

[5] Clio. *The 23-atom alphabet for σ_k palindromic cores.* Memory: `project_atom_alphabet.md`, Apr 23–26 2026.

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